

FATIMA KHALID

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SUMMARY

Computer Science graduate and AI Engineer with a strong interest in building things end to end. I've worked on AI applications involving RAG, LLMs, and agentic workflows, as well as full-stack applications using React, Next.js, and Node.js. I enjoy turning ideas into working products and figuring out how AI can be used to solve real-world problems.

EXPERIENCE

AI Lead — Datasynx (Contract)

Jul 2025 – Present

- Working as AI Lead, leading the AI engineering team for an enterprise AI agent platform, from initial architecture through to features used by client organizations.
- Handle day-to-day development of the platform's AI components alongside the team.
- Design multi-agent workflows in LangGraph, and manage sprint planning and technical documentation for the module.

AI Automation Assistant — JYC Solutions (Contract)

Jun 2025 – Apr 2026

- Worked part-time along side studies building automation workflows in n8n, connecting tools like Gemini, Airtable, Make, and Google Workspace to cut down manual work.
- Connected LLMs to external APIs, databases, and business logic to automate client workflows.
- Worked directly with clients to understand what they needed and build the automation around it.

Software Engineering Intern — Forge Internship Program, Minas Code

Jun 2024 – Aug 2024

- Co-founded a simulated startup and built a full-stack MVP with the MERN stack, Next.js, and JWT authentication.
- Added real-time features with LiveKit and built the UI with ShadCN and Zustand.
- Won both the pitch and demo competitions in the cohort.

PROJECTS

MediEase — AI-Powered Physiotherapy & Clinic Management Platform (Final Year Project)

- Building a full-stack clinic management platform for physiotherapy clinics, with an AI agent that handles appointment booking, initial diagnosis support, and prescription viewing so patients don't need to wait on staff for routine tasks.
- Built the AI agent with LangGraph and Groq LLM, handling conversational booking flow, syncing appointments to Google Calendar, and routing edge cases back to clinic staff.
- Set up the backend with PostgreSQL and Drizzle ORM, Supabase for auth and storage, and Redis with Trigger.dev for background jobs like appointment reminders.
- Developed using Next.js 15, while architecting AI agents and contributing to backend services, RAG pipelines, and overall system design.

DataSynx — AI Agent Platform for Organizations

- Leading the AI engineering side of this contract project, building an extensible AI agent that organizations can plug into their own workflows and data.
- Own the architecture for how agents are structured, coordinated, and extended for different client use cases, along with the team's workflow and documentation.
- Details of the product itself are restricted due to NDA.

Chameleon — Language Exchange Platform

- Built a language exchange platform where users can find and talk to language partners in real time.
- Built real-time chat and video calling, plus a matching system that pairs users based on the language they're learning, the language they speak, and shared interests.
- Used Next.js on the frontend and Node.js with MongoDB on the backend, focusing on keeping the matching and chat experience simple and fast.

Prompt-Based AI Video Editor

- Built a video editor where you describe the edit you want in plain text instead of using a manual timeline editor, e.g. "cut every clip where a person is speaking."
- Used OpenCV and Ultralytics for detecting objects/people in frames, and the Gemini API to interpret the prompt and decide which edit operations to apply.
- Built a Streamlit interface so the tool could actually be used end to end, from uploading a video to exporting the edited result.

SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, C++, C, SQL, HTML/CSS
Frontend: React, Next.js, Tailwind CSS, ShadCN, Zustand, EJS
Backend: Node.js, Express.js, Django REST Framework, Nest.js, RESTful API Design, JWT Authentication
AI & Agents: LangChain, LangGraph, LLMs, RAG Pipelines, Vector Databases (pgvector), Prompt Engineering, Flowise, n8n, Make, VAPI
Databases & Cloud: PostgreSQL, MongoDB, Supabase, AWS (EC2, S3), Docker, Coolify, Git/GitHub, GitHub Actions
Other: Agile/Scrum, Project Management, Technical Documentation, Team Collaboration, Problem Solving

EDUCATION

Bachelor of Science, Computer Science 2022 – 2026
COMSATS University Islamabad, Lahore Campus — CGPA: 3.46/4.0

ACTIVITIES & LEADERSHIP

- Conducted **seminars as the Co-Lead of the Artificial Intelligence Community Pakistan (AICP)** on “*Practical Applications of Large Language Models (LLMs)*”, 2025.
- Attended **workshops/seminars** on AI, distributed systems, and backend development.